

Plan for the Liquidity Agent

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Thursday, 22 May 2026



Progress

Stage	Progress
Data Ingestion	Only Live Data Gets Fetched
Is the agent able to answer queries?	—

The KPIs

Stage	Correct Claims	Answer Quality (LLM-as-a-Judge)
Development Phase 1	—	—

Sample Queries and Responses of the Agent

Computation

Step 1. Pull Last 90 Days' OHLCV Data from yfinance

- We gather 90 days worth of OHLCV data to give enough historical padding to smooth out anomalies (like a single day of massive volume due to the release of an earnings report) and calculate the stable rolling averages.
- For a daily metric $P \in \{\text{Open, High, Low, Close, Volume}\}$, one can either use the a simple moving average (SMA) or the exponential moving average (EMA).

- ① **Simple Moving Average.** It treats every day in the lookback window (say n days) with equal weight.

$$\text{SMA}_t = \frac{1}{n} \sum_{i=0}^{n-1} P_{t-i}$$

However, the problem with the SMA is that if a stock had an extreme, $10\times$ volume spike exactly 31 days ago, that spike will suddenly *drop off* the 30-day SMA calculation tomorrow. This causes the moving average to drop sharply, even though nothing actually changed in the market today.

- ② **Exponential Moving Average.** The EMA applies more weight to recent days and exponentially decays the weight of older days. It never abruptly drops data off a cliff. The EMA is calculated as:

$$\text{EMA}_t = (P_t \cdot \alpha_n) + (\text{EMA}_{t-1} \cdot (1 - \alpha_n)), \quad \alpha_n = \frac{2}{n + 1}$$

where α is called the smoothing factor. So, for a 30-day EMA, $\alpha_{30} \approx 0.0645$, i.e. the recent days get 6.5% of the weight.

- We also have to take care of the outliers. If a massive volume spike occurs (e.g. a stock trading $20\times$ its normal volume because of a meme-stock short squeeze), we can clip that day's volume to a maximum of 3 or 4 standard deviations from the historical mean. Another area of caution is that we must ensure that the OHLCV data only contains actual trading days. Gaps in dates will break rolling window logic if you use strict calendar-day maths instead of trading-day index maths.

Step 2. Compute Core Liquidity and Volatility Metrics

- We have three core metrics in this regard:

- 1 **Average Daily Volume in Dollars.** We calculate the daily dollar volume as follows:

$$\text{Daily Dollar Volume} = \text{Close Price} \times \text{Volume}$$

By calculating both a short-term (30d) and long-term (90d) moving average, our script can detect if a stock's liquidity is drying up or suddenly spiking compared to its historical norm.

- 2 **Volume Coefficient of Variation (CV).** This measures how predictable the volume is.

$$CV = \frac{\text{Standard Deviation of Volume}}{\text{Mean of Volume}}$$

A high CV means volume is erratic (dry most days, massive on others), which introduces execution risk. A low CV means steady, reliable daily trading traffic.

- 3 **Amihud (30d).** This is the *amihud illiquidity measure*. It specifically quantifies price impact (slippage potential) by calculating how much the price moves per dollar of volume.

$$\text{Amihud Ratio} = \frac{|\text{Daily Return}|}{\text{Dollar Volume}}$$

A high Amihud value means it takes very little dollar volume to violently move the stock price – a massive red flag for slippage.

Step 2.1. Average Daily Dollar Volume (ADV\$)

- Firstly, we fetch the following live data for a 90-day lookback window:

- 1 C_t : closing price for day t
- 2 H_t : high price for day t
- 3 L_t : low price for day t
- 4 V_t : volume (shares) for day t

where $t = 0, 1, 2, \dots, 90$. Note that $t = 90$ is today and $t = 0$ was 90 days ago.

- The ADV\$ measures the absolute scale of capital moving through the stock daily.
- Firstly, we calculate the daily dollar volume:

$$DV_t = C_t \times V_t$$

- Secondly, to prevent a single anomalous day from skewing the risk model, we clip any daily DV_t that exceeds the 99th percentile of the dataset to that 99th percentile value ($DV_{t,\text{clipped}}$).
- Finally, we calculate the 30-day and 90-day rolling averages, i.e. $ADV\$_k$ where $k \in \{30, 90\}$:

$$ADV\$_k = \frac{1}{k} \sum_{i=0}^{k-1} DV_{t-i,\text{clipped}}$$

If $ADV\$_{30} \ll ADV\$_{90}$, it flags a structural draining of liquidity, signaling your engine to expect higher slippage than historical baselines imply.

Step 2.2. Volume CV

- Volume CV answers: *Is the trading volume steady every day, or does the stock sit completely dead for days and then suddenly explode on a random Tuesday?*
- This is calculated as follows:

$$CV_k = \frac{\sigma_{V_k}}{\mu_{V_k}} = \frac{\sqrt{\frac{1}{k-1} \sum_{i=0}^{k-1} (V_{t-i} - \mu_{V_k})^2}}{\frac{1}{k} \sum_{i=0}^{k-1} V_{t-i}}$$

where $k \in \{30, 90\}$.

- If $CV < 0.4$, the stock is highly predictable, steady trading traffic, and consequently, safe. If $CV > 1.0$, then the stock has highly erratic liquidity. This means the average daily volume is a statistical illusion driven by a few hyperactive days. If you trade on a quiet day, your slippage will be far worse than the ADV\$ suggests.

Step 2.3. The Amihud Illiquidity Ratio

- It measures the price impact. It directly quantifies how many dollars of trading volume it takes to push the stock price by 1%. This is the rawest proxy for slippage.
- The calculation steps are as follows:

1 **Calculate Daily Absolute Return.** This is calculated as follows:

$$|R_t| = \left| \frac{C_t - C_{t-1}}{C_{t-1}} \right|$$

It is also convenient to use log returns for better mathematical properties in continuous time.

- 2 **Calculate Daily Dollar Volume.** Use the raw (unclipped) dollar volume calculated in Step 2.1, measured in millions of dollars to keep the final ratio easily readable.
- 3 **Compute the Daily Amihud Value.** Divide the absolute return by the dollar volume:

$$A_t = \frac{|R_t|}{DV_t \text{ (in Millions)}}$$

- 4 **Smooth via rolling average (30d).** Take the average of these daily ratios over a 30-day window to get a steady metric.

$$\text{Amihud (30d)} = \frac{1}{30} \sum_{i=0}^{29} A_{t-i}$$

- If the score is 0.005, then the stock is highly liquid. It takes \$100 million of volume to move the stock price by a mere 0.5%. Slippage risk is negligible. If the score is 2.500, the stock is extremely illiquid. Just \$1 million of volume will push the stock price violently by 2.5%. Attempting to execute a large order in this environment will result in massive, self-inflicted slippage.

Step 3. Structural Constraints

While Step 2 looks at activity, this step looks at the market structure to see how easily a corner can be turned on the stock.

- **Float.** The raw number of shares available to the public.
- **Short Percentage.** High short interest means a high risk of a short squeeze where liquidity completely vanishes as short sellers scramble to buy back shares at any cost.
- **Institutional Top-10 Concentration.** If the top 10 institutions own 80% of the float, those shares are functionally locked up. The effective float is much smaller than the reported float, dramatically increasing volatility potential.

Step 4. Compute Days to Liquidate at 1% and 5% Positions

- This step directly quantifies exit risk for a fund or a portfolio manager.
- If someone holds a massive position in a stock and try to sell it all at once, he will utterly destroy the price (causing massive slippage).
- To avoid this, institutional traders exit positions slowly over several days, making up only a small percentage of the market's daily volume so they blend in with regular trading traffic.
- In order to calculate days to liquidate, we need three variables:
 - 1 **Position Size (N)**. This is the total number of shares the person wants to sell.
 - 2 **Average Daily Volume (ADV)**. Computed in step 2.
 - 3 **Participation Rate (PR)**. This is the maximum percentage of daily volume one can safely represent without moving the market. In professional trading, this is typically set between 10% (i.e. 0.10) and 20% (i.e. 0.20).

With these three variables in place, DTL is calculated as:

$$\text{DTL} = \frac{\text{Position Size}}{\text{Average Daily Volume} \times \text{Participation Rate}} = \frac{N}{\text{ADV} \times \text{PR}}$$

Step 4. Compute Days to Liquidate at 1% and 5% Positions (contd.)

- Here, the DTL at 1% and 5% positions refer to a hypothetical scenario where your system tests what happens if you own 1% or 5% of the company's total float (or outstanding shares). It tries to answer the question:

If our fund takes a meaningful stake in this company, how trapped will we be if we need to get out?

- Higher DTL implies higher exit risk and vice-versa. We present the following illustration considering 10% participation rate. This illustration tries to capture the relation between liquidity, DTL and exit risk.
 - Liquid Stock.** Total Float = 10^8 shares, ADV (30d) = 5×10^6 shares/day and $N_{5\%} = \frac{5}{100} \times 10^8 = 5 \times 10^6$ shares. Therefore,

$$DTL_{5\%} = \frac{(5 \times 10^6)}{(5 \times 10^6) \times (0.10)} = 10 \text{ days}$$

This means that given 5% stake in this company, it will take us 10 days to sell off the shares without taking a financial hit.

- Illiquid Stock.** Total Float = 10^8 shares (same size as before), ADV (30d) = 2×10^6 shares/day (very dry trading traffic) and $N_{5\%} = \frac{5}{100} \times 10^8 = 5 \times 10^6$ shares. Therefore,

$$DTL_{5\%} = \frac{(5 \times 10^6)}{(2 \times 10^5) \times (0.10)} = 250 \text{ days}$$

This means that given 5% stake in this company, it will take us 250 days to sell off the shares without taking a financial hit.

Step 5. Apply Rule-Based Scoring Thresholds

We use the following rules:

Dimension	Metric	Low Risk (+0 Points)	Medium Risk (+ 1 Point)	High Risk (+2 Points)	Critical Risk (+3 Points)
Absolute Liquidity	ADV\$ (30d)	> \$10M	\$2-10M	\$500K-2M	< \$500K
Price Impact	Amihud Ratio	< 0.01	0.01-0.05	0.05-0.20	> 0.20
Volume Stability	Volume CV	< 0.3	0.3-0.6	0.6-1.0	> 1.0
Exit Capacity	DTL _{5%}	< 1 day	1-3 days	3-5 days	> 5 days
Market Structure	Free Float	> 50M shares	20-50M shares	5-20M shares	< 5M shares

Using the rules mentioned above, we generate a score for a company and categorise them into four tiers using the following table:

Category	Score	Action	Comment
Tier 1 (Unrestricted)	0-1 Points	Safe for rapid, automated market orders.	Fund can comfortably accumulate up to a 5% stake without pre-trade approval.
Tier 2 (Position Sizing Caps)	2-4 Points	The system automatically restricts maximum position size. For example, the fund is barred from owning more than 1% of the total company float.	Market orders are blocked. The trader must use Limit Orders or basic VWAP (Time-Weighted Average Price) algorithms to slowly scale in.
Tier 3 (Algorithmic Execution Only)	5-7 Points	Severe capacity constraints. The stock is flagged for extreme slippage potential.	All trades must be pushed to an institutional liquidity router or advanced execution algorithms (like VWAP with a strict 5% participation rate cap). Pre-trade compliance sign-off is required.
Tier 4 (Blacklist)	8+ Points	Automated hard-block in the trading system.	The exit door is too narrow. If the fund enters this position, Exit Risk dictates that they may become entirely trapped if the market turns.

Edge Case. When implementing this step, our rules must account for a highly dangerous structural phenomenon: the **false liquidity mirage**. Sometimes, a low-float penny stock will experience an astronomical explosion in volume (low DTL, massive ADV\$) because it is being pumped on social media or experiencing a short squeeze. If the engine only looks at recent volume, it will mistakenly score the stock as *Tier 1: Safe*. To prevent this, the step 5 logic must include a structural override:

If Float < 10M and Short% > 25%, automatically downgrade the stock by two risk tiers, no matter how high the current ADV\$ is.

This ensures that your engine recognizes that the current liquidity is temporary and artificial, protecting the portfolio from getting trapped when the pump ends.

Output

Agent's Output

- The final output of this system will manifest as a clean, high-visibility block at the top of a risk compliance dashboard, designed to give traders an immediate go/no-go signal.
- At first glance, a user will see a large, color-coded liquidity tier badge (such as a green **Tier 1. High Liquidity** or a flashing red **Tier 4. Liquidity Blacklist**) paired with a dynamic **Data Confidence Score** (e.g. **Confidence: High (100%)** or **Confidence: Poor (40%) – Data Stale**).
- Directly beneath this status block, the system will print the automated 2–3 sentence plain-English summary that explicitly cites the asset's real-world numbers. For example:

This asset is highly liquid with a 30-day ADV\$ of \$14.2M and an Amihud price impact ratio of 0.003, indicating minimal slippage risk. However, a 5% position liquidation requires 4.2 days of sustained trading volume, triggering a moderate exit risk flag.

- Below the summary narrative, the dashboard will display an interactive, scannable data grid breaking down the specific structural and activity flags that informed the final score. Each line item will feature a clear status flag alongside its raw metric, translating dense quantitative calculations into intuitive risk disclosures. For instance, if the script detects an inadequate float or an erratic volume pattern, the dashboard will display a warning label stating:

Flag Raised: Volume CV is 1.15 (Extreme Volatility) – daily trading volume is highly unpredictable, meaning realised slippage may vastly exceed the baseline model.

- Conversely, if the data pipeline encounters a data synchronisation gap, all primary metrics will be visually dimmed or accompanied by a prominent error alert reading:

Warning: Market data feed has not synced in 6 days. Liquidity scoring downgraded to Low Confidence; do not deploy capital based on these metrics.

THE END